

EV Charging Management System (CMS) – Feature Proposal

Investor Feature Report

Version: 1.0 (Pakistan Market)

Executive Summary

The proposed CMS is a cloud-based platform that enables businesses to manage EV charging stations from a single dashboard. It supports OCPP 1.6J-compatible chargers from multiple manufacturers and provides monitoring, billing, analytics, maintenance, security, and business management features.

The platform is designed for:

- Charging Network Operators (CPOs)
 - Fuel Stations
 - Shopping Malls
 - Hotels
 - Housing Societies
 - Corporate Offices
 - Fleet Operators
 - Government Organizations
 - Universities
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1. Dashboard

Live Network Overview

- Total chargers
- Online chargers
- Offline chargers
- Charging sessions in progress

- Available chargers
 - Faulted chargers
 - Reserved chargers
 - Daily revenue
 - Daily energy delivered
 - Active users
 - CO₂ emissions saved (estimated)
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Live Interactive Map

- Charger locations
 - Live status
 - Availability
 - Fault indication
 - Heatmap of charger usage
 - Geographic clustering
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2. Charger Management

Charger Registration

- Register charger
 - Assign charger ID
 - Assign operator
 - Assign station
 - QR code generation
 - RFID assignment
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Remote Operations

- Start charging remotely
- Stop charging remotely
- Restart charger
- Unlock connector
- Lock connector
- Disable charger
- Enable charger

- Emergency shutdown
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OCPP Management

- Boot Notification
 - Heartbeat monitoring
 - Meter Values
 - Status Notifications
 - Firmware Management
 - Configuration Management
 - Remote Diagnostics
 - Reset Commands
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3. Charger Health Monitoring

Real-Time Health

- Internal temperature
 - Connector temperature
 - Voltage
 - Current
 - Power
 - Cooling fan status
 - Door status
 - Communication status
 - Isolation status
 - Contactor status
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Health Score

Example

Health Score

98 / 100

✓ Communication

✓ Cooling

✓ Connector

⚠ High Internal Temperature

Recommendation:
Inspect Cooling Fan

Predictive Maintenance

Detect:

- Fan degradation
- High connector temperature
- Loose connector
- Communication instability
- Repeated restarts
- High fault frequency
- Voltage instability
- Aging charger

Maintenance recommendations generated automatically.

4. Fault Management

Automatically detect:

- Overvoltage
- Undervoltage
- Overcurrent
- Ground fault
- Emergency stop
- Communication failure
- Connector lock failure
- Isolation fault
- Internal hardware failure
- Temperature fault

Features

- Live alerts
 - SMS
 - Email
 - Push notifications
 - Maintenance ticket creation
 - Technician assignment
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5. Station Management

Each station can contain:

- Multiple chargers
 - Parking slots
 - Operator details
 - Images
 - Amenities
 - Working hours
 - Pricing policies
 - GPS location
 - Maintenance history
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6. Pricing Engine

Support:

Flat Rate

Rs. 70/kWh

Time Based

Rs. 20/min

Hybrid

Rs. 55/kWh

+

Rs. 5/min

Dynamic Pricing

- Peak hours
- Off-peak
- Weekend pricing
- Holiday pricing

Membership Pricing

Fleet Pricing

Corporate Pricing

7. Revenue Management

Daily Revenue

Monthly Revenue

Yearly Revenue

Revenue per charger

Revenue per station

Revenue per city

Revenue per operator

Tax reports

Invoice generation

Export to Excel/PDF

8. User Management

Roles

Super Admin

Operator

Station Manager

Technician

Finance

Support Agent

Permissions

Role-based access control

Audit logs

Session history

9. Fleet Management

Register companies

Assign vehicles

Assign RFID

Monthly reports

Charging cost

Energy consumption

Vehicle-wise reports

Department-wise billing

10. Reservation Management

Reserve charger

Reservation expiry

Automatic cancellation

No-show detection

Reservation analytics

11. Notifications

Charging started

Charging completed

Fault detected

Maintenance due

Firmware available

Offline charger

Revenue summary

Weekly reports

12. CCTV Integration

Connect IP cameras

Live stream

Recording

Snapshot on fault

Session-linked video

Remote viewing

AI event alerts

13. ANPR (Number Plate Recognition)

Vehicle recognition

Whitelist

Blacklist

Fleet recognition

Automatic authentication

Parking duration

Unauthorized parking detection

Automatic barrier opening

14. Smart Parking Management

Parking occupancy

Charging bay occupancy

ICE vehicle detection

Overstay detection

Parking fees

Queue monitoring

Parking reservation

15. Access Control

Barrier gates

RFID access

QR code access

PIN access

Employee cards

Visitor access

16. Solar Integration

Solar generation

Grid consumption

Battery storage

Solar utilization

Peak shaving

Energy optimization

17. Smart Energy Management

Load balancing

Demand response

Maximum load protection

Transformer protection

Power scheduling

Priority charging

Grid monitoring

18. Analytics Dashboard

Peak usage

Charging trends

Average session duration

Popular stations

Revenue trends

Failure trends

Heatmaps

Utilization %

Forecast demand

Growth analytics

19. AI Analytics

Demand prediction

Failure prediction

Revenue forecasting

Recommended maintenance

Occupancy prediction

Queue prediction

Fraud detection

Customer behavior analysis

20. Reports

Daily reports

Monthly reports

Annual reports

Energy reports

Revenue reports

Fleet reports

Maintenance reports

Fault reports

Tax reports

Custom reports

21. Maintenance Module

Maintenance scheduling

Technician assignment

Inventory management

Parts replacement

Maintenance history

Service reminders

Warranty tracking

Inspection checklists

22. Asset Management

Chargers

Power modules

Connectors

Transformers

Switchgear

Solar panels

Batteries

Warranty records

Lifecycle tracking

23. Multi-Tenant SaaS

Each customer gets:

Own logo

Own domain

Own branding

Own pricing

Own users

Own reports

Own permissions

Single backend for all customers

24. API Platform

REST APIs

Webhooks

Third-party integrations

ERP integration

Accounting software

Fleet software

Government reporting

25. Security

JWT authentication

2FA

Role-based permissions

Audit logs

Encryption

IP restrictions

Backup

Disaster recovery

26. Compliance

OCPP 1.6J

OCPP 2.0.1 Ready

OCPI Ready

OpenADR Ready

MID Meter Support

ISO 15118 Ready (future)

27. Future Roadmap

- Vehicle-to-Grid (V2G)
 - Plug & Charge (ISO 15118)
 - AI-powered charging optimization
 - Carbon credit management
 - Smart city integration
 - Battery health analytics
 - Renewable energy forecasting
 - Drone-assisted station inspection
 - Digital twin of charging stations
 - Cross-network roaming support
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Revenue Opportunities

The platform can be monetized through multiple channels:

- **SaaS subscriptions** (per charger, per station, or enterprise tier)
- **White-label licensing** for fuel companies, malls, and fleets
- **Premium AI analytics** and predictive maintenance modules
- **Maintenance and support contracts**
- **API access** for enterprise integrations
- **Custom development and deployment services**

Vision Statement

To become Pakistan's unified EV Charging Management Platform—enabling businesses to deploy, monitor, secure, and monetize EV charging infrastructure through a single intelligent, OCPP-compatible, multi-tenant cloud platform.

MVP Recommendation (Phase 1)

For an investor presentation, I'd recommend splitting features into **MVP** and **Expansion**.

MVP (6–8 months):

- OCPP 1.6J integration
- Live dashboard
- Charger management
- Remote start/stop/reset
- Health monitoring
- Fault management
- Pricing & tariffs
- Revenue dashboard
- Multi-role user management
- Reports
- Notifications

Phase 2:

- Fleet management
- CCTV integration
- ANPR
- Smart parking
- Maintenance module
- Predictive maintenance
- White-label support

Phase 3:

- Solar & energy management
- AI analytics
- V2G

- OCPI roaming
- ISO 15118 Plug & Charge
- Smart city integrations

This phased approach shows investors a credible roadmap: a practical MVP that addresses immediate market needs, followed by premium enterprise capabilities that expand the platform's value over time.